

Database management



Task_1 Library Management

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1. Introduction:

This report presents the relational structure of a comprehensive Library Management System database. The design focuses on establishing clear relationships between entities to efficiently manage books, members, loans, and related operations.

2. Core Entity Relationships:

Our database contain these tables :

AUTHORS,BOOKS,PUBLISHERS,CATEGORIES,RESERVATIONS,PENALTIES
MEMBERS,LOANS.

Relationship Matrix:

Table	Related to	Relationship to	Foreign Key	Description
Books	Authors	Many-to-one	author_id	Each book has one author
Books	Publishers	Many-to-one	publisher_id	Each book has one publisher
Books	Categories	Many-to-one	category_id	Each book belongs to one category
Loans	Books	Many_to_one	book_id	Each loan involves one book
Loans	Members	Many_to_one	member_id	Each loan is made by one member
Reservation	Books	Many_to_One	book_id	Each reservation is for one book
Reservation	Members	Many_to_One	member_id	Each reservation is by one
Penalties	Loans	One_to_One	loan_id	Each penalty stems from one loan

3. Detailed Table Relationships:

3.1 Book-Centered Relationships:

- **Authors:** One author can write multiple books (1:N)
- **Publishers:** One publisher can publish multiple books (1:N)
- **Categories:** One category can classify multiple books (1:N)

3.2 Loan Management Relationships

- **Books:** One book can be loaned multiple times (1:N)
- **Members:** One member can have multiple loans (1:N)

3.3 Penalty System Relationships

- **Loans:** One-to-one with loans (1:1)

4. Conclusion:

This relational structure provides:

- **Data integrity** through well-defined constraints
- **Operational efficiency** with optimized relationships
- **Scalability** for future expansions
- **Comprehensive tracking** of all library activities